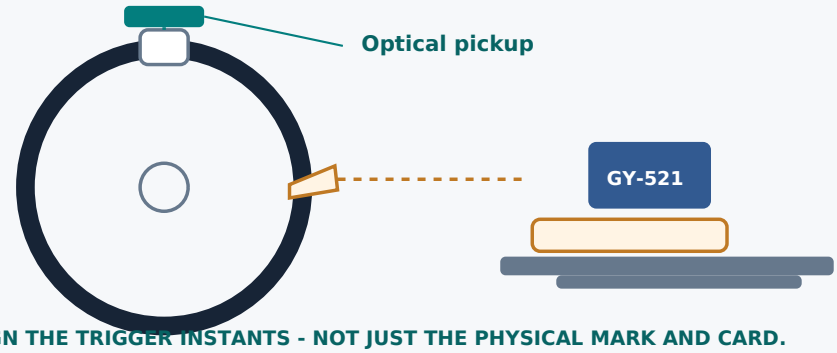
Optical-to-vibration timing calibration

Desktop wheel · single white index mark · GY-521 impulse reference

TEST RIG — CONCEPTUAL TOP VIEW



Card brushes carrier tab

A light, once-per-turn touch excites the stationary GY-521 carrier.

Thin tape / edge glue

Keep cable strain relief on the carrier; do not strike the PCB.

ALIGN THE TRIGGER INSTANTS - NOT JUST THE PHYSICAL MARK AND CARD.

PREPARE AND CONNECT

Parts: marked black wheel, optical sensor, card striker, small rigid plastic carrier/tab, firm rubber, thin double-sided tape or small edge glue dabs, GY-521.

GPIO: Optical OUT → 26; GY-521 SDA → 21, SCL → 22, INT → 27; both modules: 3V3 + common GND.

BENCH PROCEDURE

1 Mount the sensor

Fix the GY-521 to a rigid carrier with thin tape or glue at the edges. Support the carrier on firm rubber; attach a separate contact tab.

2 Establish zero angle

Turn the wheel very slowly. Identify the exact optical switching point, then position the card so it first touches the tab at that same angle.

3 Check baseline

Record 5 s stationary with the card clear. Confirm a quiet accelerometer trace and no false optical counts.

4 Record calibration passes

Record 20–30 revolutions at one steady speed. Check one optical index and one recognisable contact impulse per turn.

5 Repeat at other speeds

Repeat slow, medium and higher speeds without moving the reference positions. Log actual speed and physical setup.

6 Compare rigid mounting

Repeat with a firmer or rigid carrier mount to determine how strongly rubber compliance affects measured delay.

CALCULATE / REPORT

Δt = t(vibration onset) - t(optical edge)

Phase = 360° × Δt / revolution period

Use the first repeatable onset, not automatically the largest ringing peak. Report mean offset, variation, RPM and sample-clock fit.

PASS CHECKS AND INTERPRETATION

Check: 1 optical index per turn; 1 usable impulse per turn; no lost IMU frames or FIFO overruns; consistent offset at each speed. Retain raw optical timestamps and uncorrected acceleration samples.

Limitation: This test measures combined geometry, contact compliance, mechanical transmission and MPU-6050 filter delay. It does not by itself isolate electronic latency. Store any correction separately; do not overwrite raw measurements.